EMERSON HERRERA TOLENTINO

SOFTWARE ENGINEER · MACHINE LEARNING & GENERATIVE AI

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ACADEMIC STATEMENT

Software Engineer with more than 8 years of experience; since 2025 I have designed and improved a production RAG platform for the Central Reserve Bank of Peru. In the Master's program at PPGC – UFRGS, where I am currently a special student, I want to research how to improve and evaluate systems that combine information retrieval and LLMs: document segmentation, retrieval and answer verification. My first research experience was collaborating on a benchmark of vision-language models accepted at the LXCV workshop at CVPR 2026.

RESEARCH INTERESTS

Natural Language Processing · Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) and information retrieval · Evaluation of LLMs and RAG systems · Machine Learning · Multimodal systems · Applied AI.

EDUCATION

Special Student — Graduate Program in Computer Science (PPGC)
UFRGS, Brazil · CMP263 Machine Learning · applying to the Master's program *2026/2 — Ongoing*

Bachelor's Degree in Software Engineering
UTP — Universidad Tecnológica del Perú *2021 — 2024*

- Degree conferred on February 6, 2025.

Software Development Technician (licensed) — 3-year technical degree
SENATI *Mar 2016 — Jul 2019*

COMPLEMENTARY TRAINING

AI Engineer Specialization Program — Diploma of Excellence (19/20)
Data Science Research Perú · 92 academic hours *Mar 2026 — Aug 2026*

- RAG, agents and MCP, multimodal applications, LLM evaluation and LLMOps, with Diplomas of Excellence in five modules and a Natural Language Processing course; final project: a multi-agent system applied to business processes.

Training Program in Microservices with Java and Spring Boot
Lite Thinking · professional training course *Jun 2024 — Sep 2025*

LANGUAGES

Spanish: native · Portuguese: intermediate (good comprehension) · English: intermediate (advanced technical)

PUBLICATIONS / RESEARCH

Co-author — "Benchmark of Multimodal Vision-Language Models for Damage Detection"
LatinX in CV (LXCV) Workshop, CVPR 2026 *Jun 2026*

- Benchmark of CLIP, SigLIP, SigLIP2 and FuseLIP for natural-disaster damage detection from images and text. Work led by Mariana Risco Cosavalente, with Prof. Claudio Jung (UFRGS).
- Extended abstract accepted through double-blind peer review. My contribution: model comparison and part of the few-shot CLIP adaptation experiments on CrisisMMD and DamageDataset (github.com/emersonheto/damage_dataset_fewshot_vlms).

PROJECTS

Certilab Agentic RAG — Calibration Certificate Assistant

LangGraph, FastAPI, Qdrant, OpenAI *2026*

- Agentic RAG service combining structured and semantic search, with cited sources and access control; 7-node Adaptive RAG variant with two self-correction loops over 3,848 vectors (github.com/emersonheto/certilab-agentic-rag).

PROFESSIONAL EXPERIENCE

Analyst Programmer II | Full Stack AI Engineer / GenAI & RAG Developer

Canvia · Peru (remote) *Oct 2025 — Present*

- Client: Central Reserve Bank of Peru (BCRP). Designed, built and improved a multi-tenant RAG platform serving 4 of the bank's applications (BCRP Portal, Research Portal, NEXO 360 and Banknotes), including the ingestion and indexing of thousands of documents and hundreds of thousands of chunks; its assistant is live on the Research Portal (investigacion.bcrp.gob.pe).
- RAG pipelines for research articles and institutional documents, with multimodal ingestion of PDFs and images, vector search on Oracle 23ai, pgvector, Pinecone and Chroma, and LLMs through the OpenAI API and LangChain.

Full Stack & AI Developer (Freelance)

Certilab S.A.C. · Remoto *Sep 2024 — Present*

- Certification system in Laravel 11 and Filament with 300+ certificates issued: admin back office, client intranet to look up their own certificates, and a public QR validation portal with encrypted IDs.

Full Stack Developer | Angular, Java, .NET & Process Automation

Way Tech (WIN Empresas / Universidad Continental projects) · Peru (remote) *Nov 2024 — Oct 2025*

- Built academic and institutional platforms for Universidad Continental (24,000+ students across Huancayo, Arequipa, Cusco and Lima) with Angular and microservices in C# (.NET) and Java.

Systems Engineer

Indra · San Isidro, Peru (remote) *Mar 2021 — Aug 2024*

- Client: Telefónica del Perú. Distributed systems and microservices for banking and telecommunications environments.

IT Systems Analyst

Multilab · Pueblo Libre, Peru *Feb 2020 — Mar 2021*

- Transactional clinical systems with Laravel and CodeIgniter: new features, reporting, and roles and permissions.

Software Development Area Manager · Software Developer

MControl Energy Technology SAC · Lima, Peru *Nov 2017 — Jan 2020*

- Head of the development area (2019 — 2020) and developer (2017 — 2019): systems for service stations with C#, ASP.NET, Laravel and Vue.js.

TECHNICAL SKILLS

Generative AI & Agents: RAG (agentic and adaptive), LangChain, LangGraph, OpenAI API, Prompt Engineering, embeddings, vector databases (pgvector, Pinecone, Chroma, Qdrant), multi-agent architectures, MCP (Model Context Protocol), LLM evaluation and LLMOps

Machine Learning: Scikit-Learn, Pandas, NumPy, PyTorch, vision-language models (CLIP, SigLIP), fine-tuning and few-shot

Research: model comparison and experimentation (benchmark of vision-language models), Jupyter, LaTeX/Overleaf

Languages & Frameworks: Python (FastAPI), Java (Spring Boot), C# (.NET), Node.js, PHP (Laravel, CodeIgniter), JavaScript/TypeScript

Data & Cloud: PostgreSQL, Oracle, MySQL, MongoDB, Redis · AWS, Azure, Docker, Kubernetes, Linux